

$$\boldsymbol{H}_g = [\boldsymbol{x}_g, \tilde{\boldsymbol{x}}_g, \boldsymbol{x}_{gr}] \in \mathbb{R}^{3 \times L_f}, \quad \boldsymbol{x}_{gr} = \boldsymbol{x}_g - \tilde{\boldsymbol{x}}_g, \quad \boldsymbol{M}_g = \{\boldsymbol{H}_g, \boldsymbol{T}\}, \quad \boldsymbol{E} = \text{Emb}(\text{Tok}(\boldsymbol{D})) \in \mathbb{R}^{\gamma \times N_c \times h}$$

$$\boldsymbol{\theta}_{l_3} = \text{reshape}(\boldsymbol{E}, (\gamma, N_ch)), \quad \boldsymbol{\theta}_{\text{ts}} = \{\boldsymbol{\theta}_{l_1}, \boldsymbol{\theta}_{l_2}, \boldsymbol{\theta}_{l_3}\}$$